

Literature search report – Example project

scitech-librarian 3.2

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Item	Value
Search dates	2026-08-28 to 2026-08-28
Project	Example project
Description	the four example blocks of queries.example.json, searched twice, plus a reference list from a colleague
Sources	2 automated run(s), 1 manual source(s)
Blocks	NOV, MLP, PSC, QC
Backends / sources	openalex, arxiv, inspire, semanticscholar, ads, scopus, manual:reference-list
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Journal metrics	on file for 599 journals
Filters	sources=all

Sources

Every search that feeds this report, oldest first. 'New here' counts unique records that no earlier source had found – what each search added.

Source	Kind	Date	Method	Records	New here	Label / origin
20260828T091142	run	2026-08-28 09:11	database	5	5	first pass (OpenAlex only, NOV + MLP)
20260828T100120	run	2026-08-28 10:01	database	2931	2211	full run, six databases
reference-list	manual	2026-08-28 11:30	citation	8	2	reference list of a review article

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend's native grammar. The strings below are exactly what was sent (PRISMA-S item 8), from the most recent run of each block.

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR Thouless pump OR edge state)

Backend	Query string sent
openalex	(origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")
arxiv	(all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")
inspire	("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state")
semanticscholar	("origami" "kirigami") + ("acoustic metamaterial" "phononic crystal") + ("topological pumping" "Thouless pump" "edge state")
ads	(abs:"origami" OR abs:"kirigami") AND (abs:"acoustic metamaterial" OR abs:"phononic crystal") AND (abs:"topological pumping" OR abs:"Thouless pump" OR abs:"edge state")

Backend	Query string sent
scopus	TITLE-ABS-KEY(origami OR kirigami) AND TITLE-ABS-KEY("acoustic metamaterial" OR "phononic crystal") AND TITLE-ABS-KEY("topological pumping" OR "Thouless pump" OR "edge state")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND (amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") and ("amorphous" OR "glass" OR "disordered solid") and ("molecular dynamics" OR "structure prediction")
semanticscholar	("machine learning potential" "machine-learned interatomic potential" "neural network potential") + ("amorphous" "glass" "disordered solid") + ("molecular dynamics" "structure prediction")
ads	(abs:"machine learning potential" OR abs:"machine-learned interatomic potential" OR abs:"neural network potential") AND (abs:"amorphous" OR abs:"glass" OR abs:"disordered solid") AND (abs:"molecular dynamics" OR abs:"structure prediction")
scopus	TITLE-ABS-KEY("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND TITLE-ABS-KEY(amorphous OR glass OR "disordered solid") AND TITLE-ABS-KEY("molecular dynamics" OR "structure prediction")

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR water ingress)

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" OR "halide perovskite photovoltaic") and ("degradation" OR "stability" OR "encapsulation") and ("humidity" OR "moisture" OR "water ingress")
semanticscholar	("perovskite solar cell" "halide perovskite photovoltaic") + ("degradation" "stability" "encapsulation") + ("humidity" "moisture" "water ingress")
ads	(abs:"perovskite solar cell" OR abs:"halide perovskite photovoltaic") AND (abs:"degradation" OR abs:"stability" OR abs:"encapsulation") AND (abs:"humidity" OR abs:"moisture" OR abs:"water ingress")
scopus	TITLE-ABS-KEY("perovskite solar cell" OR "halide perovskite photovoltaic") AND TITLE-ABS-KEY(degradation OR stability OR encapsulation) AND TITLE-ABS-KEY(humidity OR moisture OR "water ingress")

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization OR critical states OR fractal spectrum)

arXiv receives groups [0, 2] only (nested-boolean limitation).

Backend	Query string sent
openalex	(quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")
arxiv	(all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")
inspire	("quasicrystal" OR "quasiperiodic lattice") and ("photonic" OR "waveguide" OR "optical lattice") and ("localization" OR "critical states" OR "fractal spectrum")

Backend	Query string sent
semanticscholar	("quasicrystal" "quasiperiodic lattice") + ("photonic" "waveguide" "optical lattice") + ("localization" "critical states" "fractal spectrum")
ads	(abs:"quasicrystal" OR abs:"quasiperiodic lattice") AND (abs:"photonic" OR abs:"waveguide" OR abs:"optical lattice") AND (abs:"localization" OR abs:"critical states" OR abs:"fractal spectrum")
scopus	TITLE-ABS-KEY(quasicrystal OR "quasiperiodic lattice") AND TITLE-ABS-KEY(photonic OR waveguide OR "optical lattice") AND TITLE-ABS-KEY(localization OR "critical states" OR "fractal spectrum")

Results summary

Block	openalex	arxiv	inspire	semanticscholar	ads	scopus	manual:reference-list	Identified	Retrieved	Unique
NOV	0	0	0	0	0	0	-	0	0	0
MLP	478	138	0	109	203	149	8	1,085	815	446
PSC	4567	175	0	2968	2179	4731	-	14,620	1275	1163
QC	154	414	17	122	206	65	-	978	854	609
Total	5,199	727	17	3,199	2,588	4,945	8	16,683	2944	2218

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs); summed over runs, and for manual sources the number of records ingested. Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

Timeline

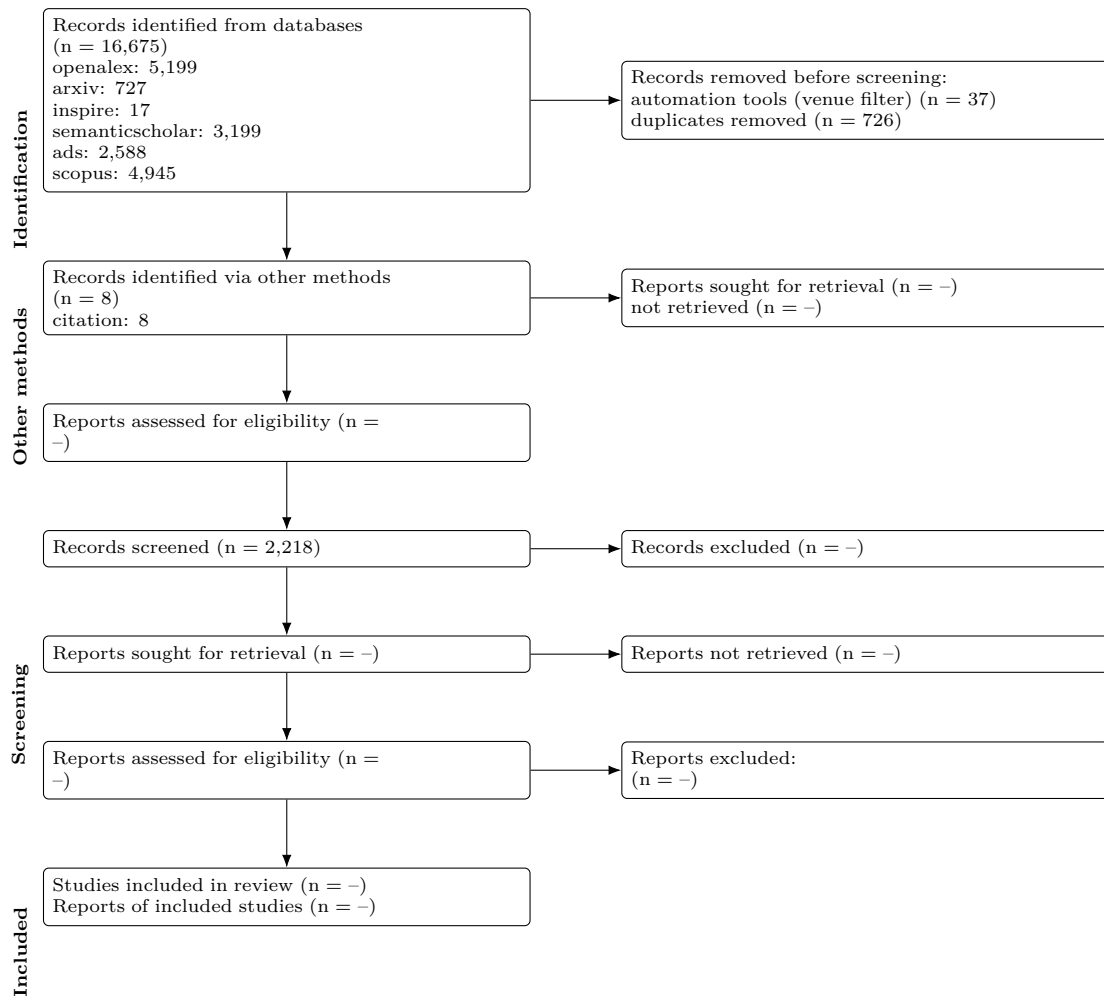
Per-block hit totals in each automated run (sum over backends), oldest first; drift shows how the indexes – or the queries – changed.

Block	20260828T091142	20260828T100120
NOV	0	0
MLP	239	838
PSC	-	14,620
QC	-	978

When records entered the project

Month	Records first seen
2026-08	2218

PRISMA 2020 flow



Stage	n
Records identified from databases	16,675
openalex	5,199
arxiv	727
inspire	17
semanticscholar	3,199
ads	2,588
scopus	4,945
Records identified via other methods	8
citation	8
Records retrieved (downloaded / ingested)	2,981
Removed before screening: automation (non-curated venues)	37
Removed before screening: duplicates	726
Records to screen (unique)	2,218
Records screened	2,218 (assumed = unique)
Records excluded at screening	-
Reports sought for retrieval	-
Reports not retrieved	-
Reports assessed for eligibility	-
Other methods: reports sought	-
Other methods: reports not retrieved	-
Other methods: reports assessed	-
Studies included	-
Reports of included studies	-

Automation stages are computed from the data; '-' marks manual stages not yet recorded in screening.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within --limit, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire, semanticscholar, ads, scopus (documented public APIs)
2	Multi-database searching	6 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	none recorded
5	Citation searching	reference-list
6	Contacts	none recorded
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json of each run
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied; report filters: sources=all
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	2 run(s) combined; see Timeline
13	Dates of searches	searched on 2026-08-28 to 2026-08-28
14	Peer review	none
15	Total records	16,675 identified from databases, 8 via other methods; 2,981 retrieved; 2,218 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 726 duplicates removed

Top 10 records per block

Deduplicated across sources, sorted by cited. The complete set is in all_records.csv / .ris of each run.

Block MLP (446 unique)

Title	Authors	Year	Venue	Cited	DOI	Scopus CiteScore
Atom-centered symmetry functions for constructing high-dimensional neural network potentials	Jörg Behler	2011	The Journal of Chemical Physics	1643	10.1063/1.3553717	17.3 (2025)
Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	The Journal of Physical Chemistry Letters	269	10.1021/acs.jpclett.8b00902	18.8 (2025)
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2020	npj Computational Materials	195	10.1038/s41524-020-00367-7	17.4 (2025)
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2019	Cambridge University Engineering Department Publications Database	184	10.17863/cam.55408	
Growth Mechanism and Origin of High σ Content in Tetrahedral Amorphous Carbon	Caro, Miguel A.; Deringer, Volker L.; Koskinen, Jari et al.	2018	Phys. Rev. Lett.	120, 178	10.1103/PhysRevLett.120.166101	166101 (2018)
Constructing first-principles phase diagrams of amorphous Li_xSi using machine-learning-assisted sampling with an evolutionary algorithm	Artrith, Nongnuch; Urban, Alexander; Ceder, Gerbrand	2018	The Journal of Chemical Physics	155	10.1063/1.5017661	
Study of Li atom diffusion in amorphous Li_3PO_4 with neural network potential	Wenwen Li; Yasunobu Ando; Emi Minamitani et al.	2017	The Journal of Chemical Physics	154	10.1063/1.4997243	17.3 (2025)
Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases	Taiping Hu; Jianxin Tian; Fuzhi Dai et al.	2022	Journal of the American Chemical Society	107	10.1021/jacs.2c11321	17.2 (2025)
Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon	Qian, X.; Peng, S.; Li, X. et al.	2019	Materials Today Physics	103	10.1016/j.mtphys.2019.100140	17.9 (2025)
A unified deep neural network potential capable of predicting thermal conductivity of silicon in different phases	Li, R.; Lee, E.; Luo, T.	2020	Materials Today Physics	102	10.1016/j.mtphys.2020.100181	17.0 (2025)

Block PSC (1163 unique)

Title	Authors	Year	Venue	Cited	DOI	Scopus CiteScore
Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance	Michael Saliba; Taisuke Matsui; Konrad Domanski et al.	2016	Science	3691	10.1126/science.aah5557	
High-efficiency two-dimensional Ruddlesden–Popper perovskite solar cells	Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.	2016	Nature	3360	10.1038/nature18306	
Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)	Jung, Eui Hyuk; Jeon, Nam Joong; Park, Eun Young et al.	2019	Nature	2318	10.1038/s41586-019-1036-3	
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie International Edition	2006	10.1002/anie.201405466	46.766 (2025)
Review of recent progress in chemical stability of perovskite solar cells	Guangda Niu; Xudong Guo; Liduo Wang	2014	Journal of Materials Chemistry A	1942	10.1039/c4ta04991a	11.5 (2025)
Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics	Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.	2018	Chemical Reviews	1767	10.1021/acs.chemrev.8b00336	109.3 (2025)
Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell	Lee, Jin-Wook; Kim, Deok-Hwan; Kim, Hui-Seon et al.	2015	Advanced Energy Materials	1661	10.1002/aenm.201500310	35.9 (2025)
23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability	Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.	2017	Nature Energy	1533	10.1038/nenergy.2017.9	76.7 (2025)
Investigation of CH ₃ NH ₃ PbI ₃ Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques	Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.	2015	ACS Nano	1342	10.1021/nn506864b	23.1 (2025)
Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells	Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al.	2014	Nano Letters	1198	10.1021/nl501982b	13.5 (2025)

Block QC (609 unique)

Title	Authors	Year	Venue	Cited	DOI	Scopus CiteScore
Transport and Anderson localization in disordered two-dimensional photonic lattices	Schwartz, Tal; Bartal, Guy; Fishman, Shmuel et al.	2007		1329	10.1038/nature05623	
Observation of two-dimensional discrete solitons in optically induced nonlinear photonic lattices	Fleischer, Jason W.; Segev, Mordechai; Efremidis, Nikolaos K. et al.	2003		1039	10.1038/nature01452	
Hyperuniform states of matter	Salvatore Torquato	2018	Physics Reports	501	10.1016/j.physrep.2018.03.001	518.4 (2025)
Localization and delocalization of light in photonic moiré lattices	Wang, Peng; Zheng, Yuanlin; Chen, Xianfeng et al.	2020	Nature	577, 482	10.1038/s41586-019-1851-6	
Decoherence and Disorder in Quantum Walks: From Ballistic Spread to Localization	Schreiber, A.; Cassemiro, K. N.; Potoček, V. et al.	2011		355	10.1103/PhysRevLett.106.180403	
Delocalization of a disordered bosonic system by repulsive interactions	B. Deissler; Matteo Zaccanti; G. Roati et al.	2010	Nature Physics	273	10.1038/nphys1635	29.4 (2025)
Topological triple phase transition in non-Hermitian Floquet quasicrystals	Weidemann, Sebastian; Kremer, Mark; Longhi, Stefano et al.	2022	Nature	223	10.1038/s41586-021-04253-0	
Observation of topological valley modes in an elastic hexagonal lattice	Vila, Javier; Pal, Raj Kumar; Ruzzene, Massimo	2017		219	10.1103/PhysRevB.96.134307	
Disorder-Enhanced Transport in Photonic Quasicrystals	Liad Levi; Mikael C. Rechtsman; Barak Freedman et al.	2011	Science	205	10.1126/science.1202977	
Compact localized states and flat-band generators in one dimension	Maimaiti, Wulayimu; Andreanov, Alexei; Park, Hee Chul et al.	2017		184	10.1103/PhysRevB.95.115135	

Suggestions

- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block PSC: openalex 4,567, semanticscholar 2,968, ads 2,179, scopus 4,731 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,731); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.

- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow’s manual stages are empty: fill in `screening.json` (screened / excluded / assessed / included) and rerun `report.py` to complete the diagram.